

SUNWAY UNIVERSITY



***Factors Influencing Customer Expenditure in E-Commerce: A
Statistical Analysis Using R***

Individual Assignment

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1.0 Introduction

E-commerce has reshaped how consumers shop and, just as importantly, how much of that shopping now leaves a data trail. Every checkout logs quantity, price, discount, category, and customer history; these are valuable materials that businesses can use to understand what actually drives spending rather than guessing at it. As competition intensifies across online retail, this shift from intuition to evidence has become central to pricing, targeting, and revenue strategy.

Turkey is a useful setting for this question because its e-commerce sector is expanding unusually fast. Figures from the Turkish Ministry of Trade's Electronic Commerce Information System (ETBİS) put 2025 e-commerce volume at roughly 4.57 trillion Turkish Lira (about USD 115.4 billion), which is a 52.2% increase on the prior year, across more than 5.94 billion transactions. That scale of activity is what makes transaction-level data from this market genuinely informative rather than a small, noisy sample.

Most existing research, however, stops short of transaction-level spending. Studies of Turkish and international online shoppers tend to focus on adoption, purchase intention, and channel attitudes, and typically examine demographic, behavioural, or promotional factors in isolation rather than together. As a result, businesses still face challenges in identifying which factors most strongly influence spending and how to use them to support data-driven decision-making.

Therefore, this study investigates the factors influencing customer expenditure in an e-commerce environment using transaction-level data. The findings are expected to provide practical insights for e-commerce businesses seeking to improve pricing strategies, promotional effectiveness, customer targeting, and revenue generation.

1.1 Research Questions

RQ1: Does order quantity significantly influence total customer expenditure?

RQ2: Does unit price significantly influence total customer expenditure?

RQ3: Does customer age significantly influence total customer expenditure?

RQ4: Does discount amount significantly influence total customer expenditure?

RQ5: How does customer expenditure vary across different product categories?

RQ6: Does customer type (returning vs. new customers) significantly influence total customer expenditure?

RQ7: Which factors are the strongest predictors of customer expenditure?

1.2 Research Objectives

1. To examine the relationship between order quantity and total customer expenditure.
2. To investigate the effect of unit price on total customer expenditure.
3. To determine whether customer age significantly influences total customer expenditure.
4. To assess the impact of the discount amount on total customer expenditure.
5. To investigate whether total customer expenditure differs significantly between returning and new customers.
6. To analyze the impact of product category classification on total transaction expenditure.
7. To develop and evaluate regression models that explain customer spending behaviour.

2.0 Literature Review

The growth of e-commerce has encouraged researchers to investigate the factors influencing online purchasing behaviour and customer spending. While early studies focused primarily on technology adoption and online shopping intentions, more recent research has shifted toward understanding the drivers of actual purchase behaviour and expenditure. Existing literature suggests that customer spending is influenced by a

combination of demographic characteristics, purchasing behaviour, pricing strategies, and promotional activities.

Research examining online shopping behaviour in Turkey highlights the importance of demographic characteristics in shaping consumer participation in e-commerce. Çebi Karaaslan, K.(2022), using data from more than 47,000 Turkish households, found that younger consumers and individuals with higher education and income levels were more likely to engage in online shopping activities. Altundag (2025) similarly reported that consumers aged 18 to 24 were the most active e-commerce users in Türkiye, while participation declined among older age groups due to differences in digital literacy and technology adoption. These findings suggest that age may influence consumer purchasing patterns and expenditure levels, although the strength of this relationship remains inconsistent across studies.

Beyond demographic characteristics, purchasing behaviour has been consistently identified as a major determinant of spending. Kooti et al. (2016), analysing more than 121 million online purchases, observed that consumers who purchased more items generally spent more overall, indicating a strong relationship between purchase quantity and total expenditure. Their findings further revealed that spending behaviour varies across demographic groups and tends to peak among middle-aged consumers before declining in older age groups. Such evidence suggests that transaction-level behavioural variables may provide stronger explanations of spending than demographic characteristics alone.

Pricing-related factors complicate this further. Standard theory suggests higher prices raise transaction value when quantities are held constant, but promotional savings do not behave in a simple, always-positive way. Stilley, Inman, and Wakefield (2010), in a field study of shoppers, found that discounts on planned purchases increased spending only under specific budget conditions; savings encountered while a shopper still had unallocated budget were channelled into stockpiling, while savings encountered after that budget was exhausted spilled into unplanned purchases instead. Discount effectiveness, in other words, depends on context, not just size.

The effectiveness of promotional strategies remains a widely debated topic in e-commerce research. Al Mutanafisa and Retnaningsih (2021) reported that sales promotions significantly increased impulsive online purchases. This finding supports the view that promotional incentives can stimulate purchasing activity. Collectively, these studies indicate that promotional variables may influence customer expenditure, but the direction and magnitude of the effect can vary across situations.

In addition to transactional drivers, consumer expenditure patterns differ depending on the product being purchased. While early e-commerce literature evaluated digital retail as a single homogenous sales channel, Knezevic et al. (2021) demonstrated that consumer attitudes, search practices, and shopping behaviors vary significantly across distinct product categories, such as clothing and footwear, consumer electronics, and groceries. Specifically, high-involvement verticals like consumer electronics drive higher research efforts and larger baseline transaction sizes compared to lower-cost, high-frequency categories like groceries. Accounting for product category differentiation is therefore critical to understanding variance in customer spending.

Finally, researchers have increasingly recognized the importance of post-purchase behaviour. Röllecke et al. (2017) argued that customer-friendly return policies can substantially increase customer lifetime value and future spending. Their findings indicate that returning customers often generate higher long-term profitability, highlighting the importance of maintaining positive customer relationships beyond the initial purchase transaction.

Although previous studies provide valuable insights into online consumer behaviour, several gaps remain. First, many studies focus on purchase intention, technology adoption, or online shopping participation rather than actual expenditure. Second, existing research frequently examines demographic, behavioural, or promotional factors separately rather than integrating them within a single analytical model. Third, evidence regarding the effects of age and discounts on spending remains mixed, suggesting the need for further empirical investigation using transaction-level data. Therefore, this study seeks to address these gaps by examining how order quantity, unit price,

customer age, product category, and discount amount influence total customer expenditure using e-commerce transaction data.

2.1 Hypothesis Development

Based on the literature reviewed and the objectives of the study, the following hypotheses are proposed:

H1: Order quantity has a significant relationship with total customer expenditure.

H2: Unit price has a significant relationship with total customer expenditure.

H3: Customer age has a significant relationship with total customer expenditure.

H4: Discount amount has a significant relationship with total customer expenditure.

H5: Total customer expenditure varies significantly across product categories.

H6: Returning customers spend significantly differently from new customers.

3.0 Methodology

3.1 Research Design

This study adopts a quantitative research design to investigate the factors influencing customer expenditure in an e-commerce environment. The analysis is based on transaction-level data and focuses on identifying statistically significant determinants of spending behaviour through descriptive statistics, exploratory data analysis, hypothesis testing, and regression modelling. The entire analysis was conducted using R, which provides extensive capabilities for data preparation, statistical testing, visualisation, and predictive modelling.

3.2 Dataset Description

The analysis uses the E-Commerce Customer Behavior and Sales Dataset obtained from Kaggle. The dataset consists of 5,000 transaction records collected from a Turkish e-commerce platform between January 2023 and March 2024. Each record represents an individual purchase transaction and contains information relating to customer demographics, purchasing behaviour, pricing characteristics, product attributes, and transaction outcomes.

The dataset comprises a mixture of 3 numerical, 6 integer, 8 categorical, and 1 date variable. The diversity of variables available within the dataset provides an opportunity to examine customer expenditure from multiple perspectives. Specifically, the dataset allows the simultaneous investigation of behavioural factors, pricing decisions, demographic characteristics, and customer retention status, making it appropriate for addressing the research objectives of this study.

3.3 Data Preparation and Cleaning

Before conducting statistical analysis, data preparation and cleaning were done to ensure the reliability and suitability of the dataset. The dataset structure, variable types, summary statistics, missing values, and duplicate observations were examined using R.

The dataset contained zero missing values and zero duplicate records, so no imputation or record removal was necessary. To improve consistency throughout the analysis, variable names were standardised using the `clean_names()` function from the `janitor` package. Date variables were converted into date format, while categorical variables were transformed into factor variables to ensure compatibility with statistical testing and regression modelling procedures.

Following data standardisation, outlier detection was conducted using the Interquartile Range (IQR) method across all numerical variables. This approach enabled the identification of unusually high or low observations that could potentially influence statistical estimates. Although several observations exceeded the IQR thresholds, these

values were retained because they could represent realistic customer transactions rather than obvious data-entry errors. In an e-commerce environment, extreme purchase values may reflect genuine customer behaviour and can be an important component of business performance. Hence, retaining these observations was considered more appropriate than removing them from the analysis.

3.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was undertaken to develop an understanding of the dataset before proceeding with statistical modelling and making analytical decisions. Measures including the mean, median, standard deviation, interquartile range, and skewness were examined to evaluate central tendency, dispersion, and distributional characteristics. These measures provided an initial indication of the variability and distribution of customer expenditure and other explanatory variables.

Graphical analysis was conducted using histograms, scatterplots, boxplots, and correlation matrices. Histograms were used to assess the distributional properties of numerical variables, customer expenditure, and discount amount. Scatterplots were employed to examine the relationships between expenditure and continuous predictors such as quantity purchased, discount amount, and unit price, while boxplots were used to compare expenditure across different product categories. A correlation matrix was also generated to assess the strength and direction of relationships among numerical variables and to identify potential multicollinearity concerns before model estimation.

3.5 Data Transformation

The results of the exploratory analysis suggested that the distribution of customer expenditure (total_amount) turned out to be sharply right-skewed, which showed that a small number of high-value orders were pulling the mean well above the median (*Appendix Figure 1*). Such behaviour is common in transaction-level retail datasets, where a relatively small proportion of customers generate disproportionately large purchases.

To reduce skewness and improve the suitability of the dependent variable for regression analysis, a natural logarithm transformation was applied to Total Amount. The transformed variable was retained alongside the original expenditure measure, allowing the performance of models estimated on the original scale and transformed scale to be directly compared (*Appendix Figure 1*).

This approach was adopted because transformation can improve the validity of statistical inference by reducing heteroscedasticity and improving the distributional properties of model residuals. Rather than assuming that transformation would automatically improve model performance, both specifications were evaluated during the model comparison stage.

3.6 Statistical Modelling Strategy

The analytical framework was designed to address two distinct aspects of customer expenditure behaviour. The first objective was to identify the factors that influence transaction expenditure, while the second objective was to determine whether expenditure differs between returning and new customers. Since these objectives involve different statistical questions, separate analytical techniques were employed. To investigate the determinants of customer expenditure, a progressive regression modelling strategy was adopted. Four regression models were estimated, with each model introducing additional explanatory variables to provide a more comprehensive understanding of spending behaviour.

The first model employed simple linear regression and evaluated quantity alone as a baseline. The second model focused specifically on pricing-related influences by incorporating unit price and discount amount. These variables are directly controllable by e-commerce managers and therefore provide practical insights into pricing and promotional strategies. The third model extended the analysis by integrating behavioural, demographic, and product-related variables simultaneously. This specification enabled expenditure to be examined within a broader decision-making context while controlling for multiple factors that may influence purchasing behaviour.

Finally, a fourth model was estimated using the same predictors as the third model but with log-transformed expenditure as the dependent variable. This model was developed to determine whether a transformed specification could improve model fit and better satisfy the assumptions of linear regression.

In the third and fourth models, the variable product category is a factor variable; R dummy-encodes it into $k-1$ binary indicators against an omitted baseline category. As each dummy coefficient only tests significance against that one baseline, a Type-II ANOVA is used separately to test H5, whether category matters as a whole, not just relative to whichever category happened to be dropped.

In addition to this analysis, a Welch Two-Sample t-test was conducted to evaluate whether average expenditure differs significantly between returning and new customers. This analysis was performed separately because returning customer status represents a binary grouping variable rather than a continuous predictor. Hence, comparing group means through hypothesis testing was considered a more direct and interpretable approach for addressing this research question.

3.7 Model Evaluation and Diagnostic Procedures

Model evaluation was conducted using a combination of goodness-of-fit measures and diagnostic tests. The performance of competing regression models was assessed through comparisons of R^2 , Adjusted R^2 , and Root Mean Squared Error (RMSE). To assess multicollinearity among predictor variables, Variance Inflation Factor (VIF) statistics were calculated. This procedure was necessary because highly correlated explanatory variables can inflate standard errors and reduce the reliability of coefficient estimates. Evaluating VIF values therefore ensured that the estimated effects of individual predictors remained stable and interpretable.

Residual diagnostic analysis was also performed for both the raw-scale and log-transformed regression models. Diagnostic plots were examined to assess linearity, homoscedasticity, normality of residuals, and the presence of influential observations. Comparing the diagnostic performance of the two models provided additional evidence

regarding whether the logarithmic transformation improved compliance with regression assumptions.

The model demonstrating the strongest overall explanatory performance while satisfying the underlying assumptions of linear regression was selected as the final model for interpretation.

4.0 Analysis & Results

This section presents the key findings from the analysis of customer expenditure. It begins with the main patterns identified through exploratory analysis, followed by regression analysis, hypothesis testing, and model validation.

4.1 Exploratory Findings

The exploratory findings reveal considerable variation in customer expenditure across the characteristics examined. In particular, differences were evident across product categories, unit prices, and purchase quantities, while demographic characteristics showed comparatively weaker patterns.

The age distribution of customers appeared relatively balanced across the observed range (*Appendix Figure 2*). Unlike expenditure, customer age did not exhibit significant skewness or concentration within a specific age group. This suggests that the dataset represents a reasonably diverse customer population and reduces concerns that findings may be driven by a narrow demographic segment.

The differences in spending behaviour were also observed across product categories (*Appendix Figure 3 and Appendix Table 1*). Electronics recorded the highest average expenditure and displayed the widest variation in transaction values, whereas categories such as Food, Books and Beauty generated lower spending levels. These differences indicate that expenditure patterns vary considerably across product categories and suggest that product type may play an important role in explaining

transactions. Hence, product category was retained as an explanatory variable in the regression models.

The bivariate analyses provided further insight into the factors associated with customer expenditure. A positive relationship was observed between unit price and expenditure (*Appendix Figure 4*), indicating that customers purchasing higher-priced products generally generated larger transaction values. However, variability increased as unit prices rose, suggesting that pricing alone does not fully explain spending behaviour and that other transaction characteristics may also influence expenditure.

Among all variables examined during the exploratory stage, purchase quantity demonstrated the strongest and most consistent relationship with expenditure (*Appendix Figure 5*). Transaction values increased steadily as the number of purchased items increased, with relatively limited deviation from the fitted trend line. This finding provided evidence that quantity would likely emerge as a significant predictor of expenditure and offered early support for Hypothesis H1.

Discount amount needed its own look before joining the correlation matrix, because its distribution is a different shape entirely from the other numeric variables: 69.5% of orders in this dataset carry no discount at all (*Appendix Figure 6*). Among the remaining orders, discount size is right-skewed in the same direction as total amount and unit price, but far more concentrated at zero. The scatterplot produces a result that looks contradictory at first: the raw correlation between discount amount and total amount is positive ($r = 0.348$), not negative. Bigger discounts go with bigger orders, not smaller ones, because discount size tracks unit price and quantity directly, rather than working against them.

The correlation analysis further reinforced these observations (*Appendix Figure 7*). Quantity and unit price exhibited the strongest positive correlation with expenditure, whereas variables such as age and session duration demonstrated comparatively weak relationships. These findings suggest that transactional characteristics are likely to

contribute more to expenditure variation than demographic characteristics. Overall, the exploratory analysis identified clear variation in expenditure across transactional characteristics and provided an initial basis for the subsequent hypothesis tests and regression modelling. The formal statistical tests were then used to determine whether the observed patterns were statistically significant.

4.2 Regression Analysis

The regression findings show that when more predictors are added, the models' explanatory power gradually increases (*Appendix Table 2*). Model 1, which examines quantity alone, explains 10.83% of the variation in customer expenditure. Although the relationship is statistically significant, its relatively low explanatory power indicates that quantity alone provides only a limited explanation of differences in spending. This is consistent with the positive but comparatively modest bivariate relationship observed in the exploratory analysis.

The inclusion of unit price and discount amount in Model 2 significantly increases the explanatory power of the model, with R^2 rising to 62.59%. This represents the largest improvement between successive models and highlights the importance of pricing-related variables in explaining transaction expenditure. Model 3 further includes demographic and product-related characteristics, increasing the explanatory power to 73.66%. This improvement indicates that customer expenditure is influenced by a combination of purchasing, pricing, demographic, and product-related factors rather than by any single variable in isolation.

Model 4, which applies the log-transformed expenditure measure while retaining the predictors from Model 3, provides the strongest overall performance. The model achieves an R^2 of 0.8551 and an adjusted R^2 of 0.8548, indicating that approximately 85.5% of the variation in customer expenditure is explained by the variables included in the model. Therefore, Model 4 was selected as the final model for interpretation and hypothesis testing.

Overall, the progression across the models demonstrates that transaction-level and product-related characteristics significantly improve the explanation of customer expenditure. The relatively small improvement associated with demographic information also suggests that customer characteristics such as age may provide less additional explanatory value once transaction and product factors are taken into account. The final model therefore provides a strong basis for examining the individual relationships proposed in the hypotheses.

4.3 Hypothesis Testing

The final log-transformed regression model was used to evaluate Hypotheses H1 to H6 (*Appendix Table 3*). Purchase quantity (H1) and unit price (H2) are both significant, positive predictors of customer expenditure, confirming the expected relationships. Purchase quantity represents the strongest continuous predictor in the model, indicating that larger basket sizes are associated with higher transaction values, while higher unit prices are likewise associated with greater expenditure.

Customer age (H3) is not statistically significant ($p = 0.7407$), with a negligible coefficient indicating that age provides little additional explanatory value once transaction and product characteristics are taken into account. This suggests that age-based targeting would not meaningfully predict order value within this dataset. Discount amount (H4) is statistically significant but negatively associated with expenditure. However, this result should be interpreted carefully as an observed association rather than evidence that discounts cause customers to spend less. A likely explanation is that larger discounts are concentrated in lower-value orders or lower-priced product categories within the dataset, meaning that the negative coefficient may reflect where discounting occurs rather than its causal effect on individual customer spending.

Product category (H5) is statistically significant overall based on the Type-II ANOVA, even after controlling for price, quantity, and discount. This indicates that differences in transaction value across categories cannot be explained solely by what customers

purchase or how much they pay, suggesting that product category represents an additional structural factor in expenditure.

Hypothesis H6 examined whether returning and new customers differed significantly in average expenditure per transaction and was assessed using a Welch Two-Sample t-test. The results show that the difference between the two groups is not statistically significant ($p = 0.899$), indicating that returning and new customers spend statistically indistinguishable amounts per transaction. This does not imply that customer retention lacks business value; rather, it suggests that repeat-customer status is not itself predictive of transaction size. The potential value of retention may instead be reflected in purchase frequency over time, which cannot be assessed from the current transaction-level analysis.

Overall, the hypothesis testing results indicate that transaction and product characteristics, particularly purchase quantity, unit price, discount amount, and product category, are more closely associated with customer expenditure than demographic characteristics or customer status. The findings therefore suggest that spending behaviour is primarily shaped by what customers purchase and the characteristics of those transactions rather than whether the customer is new or returning.

4.4 Model Validation

Variance Inflation Factors for all predictors in Models 3 and 4 came in well under the conventional threshold of 5, so multicollinearity is not a concern (*Appendix Table 4*). Residual diagnostics show better explanation: the raw-scale model (*Appendix Figure 8*) shows a funnel shape in the residuals-vs-fitted plot and a Q-Q line that bows away sharply in the upper tail, which shows heteroscedastic, non-normal residuals, exactly what the skewed outcome variable would predict. The log-transformed model (*Appendix Figure 9*) is markedly better across most of the fitted-value range, though a handful of very high-value transactions still show mild fanning at the end. That is an honest improvement rather than a perfect fix, but enough to justify treating Model 4 as the model of record for interpretation.

5.0 Discussion and Managerial Implications

Based on the findings, several practical recommendations can be proposed for e-commerce managers.

First, businesses should focus on strategies that encourage customers to purchase more items per order. Product bundling, volume discounts, cross-selling recommendations, and minimum-spend incentives may help increase average order value by leveraging the strong positive relationship between quantity and expenditure.

Second, product categories associated with higher spending should receive greater strategic attention. Premium products and higher-value categories may contribute disproportionately to revenue generation and should therefore be prioritised in marketing campaigns and inventory planning.

Third, discount strategies should be reviewed carefully. The results suggest that larger discounts do not necessarily translate into higher expenditure. Rather than offering broad discounts across all products, firms may benefit from targeted promotional campaigns designed to increase basket size while preserving profit margins.

Finally, because age and returning-customer status were not significant predictors of spending, businesses should avoid relying solely on demographic segmentation when attempting to increase revenue. Behavioural indicators such as purchasing patterns and transaction characteristics appear to provide more useful insights for decision-making.

6.0 Conclusion

This study modelled customer expenditure in a 5,000-transaction e-commerce dataset using a progressive regression strategy in R, moving from a single-predictor baseline to a full, log-transformed model. Quantity, unit price, discount amount, and product category all emerged as significant predictors of spend; customer age and returning-customer status did not. Quantity was the strongest single driver, with each additional unit associated with roughly a 55.5% increase in expenditure.

The log-transformed multiple regression model produced the strongest performance, achieving the highest explanatory power ($R^2 = 0.8551$) while also providing better compliance with the assumptions of linear regression. Diagnostic testing confirmed that issues related to skewness and heteroscedasticity were significantly reduced following the log transformation.

The main limitation is that this is a single secondary dataset from one platform over a 14-month window; it describes what predicted spend in this sample, not what will hold everywhere. A natural extension would be a classification model, such as logistic regression, to test more formally whether transaction-level variables can distinguish returning from new customers, since the t-test used here only establishes that their average spend does not differ. Overall, the evidence points in the same direction throughout: expenditure in this dataset is driven by what customers buy and how much of it, not by who they are demographically. For e-commerce firms, that is a direct argument for building revenue strategy around basket size and category mix rather than demographic targeting.

7.0 References

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8.0 Appendix

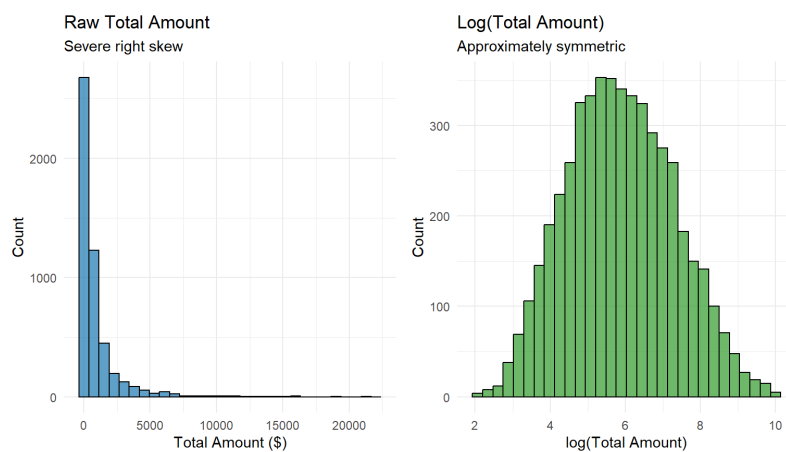


Figure 1: Distribution of Raw and Log Total Customer Expenditure

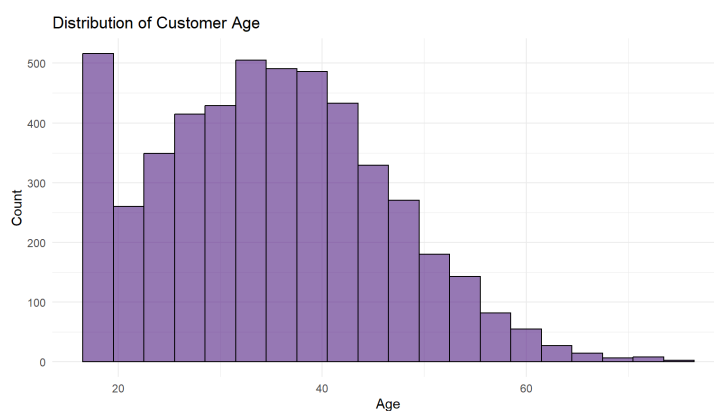


Figure 2: Distribution of Customer Age

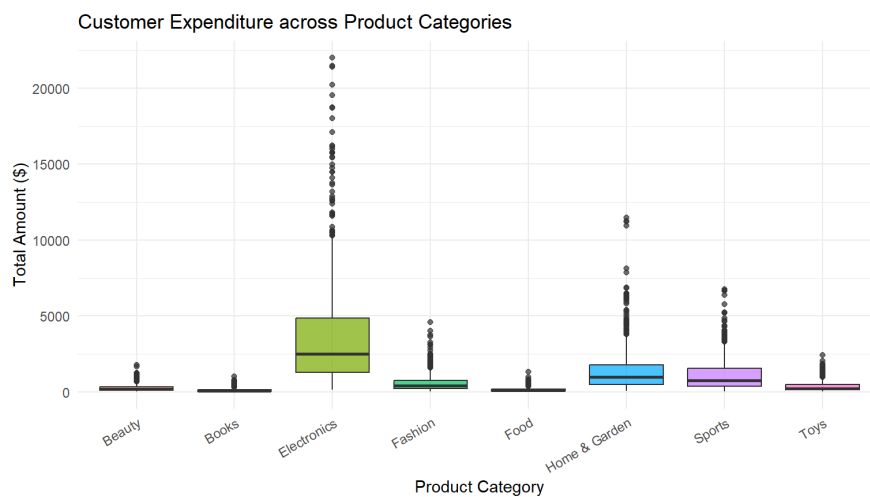


Figure 3: Customer Expenditure by Product Category

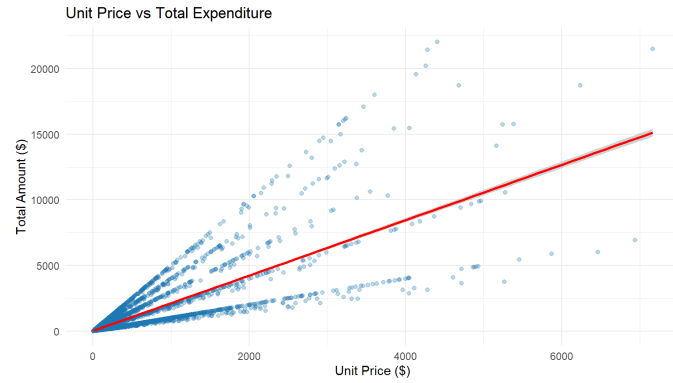


Figure 4: Unit Price versus Total Expenditure

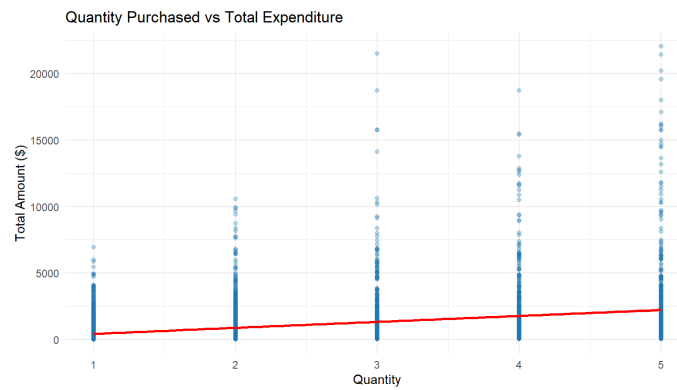


Figure 5: Quantity Purchased versus Total Expenditure



Figure 6: Distribution of Discount Amount and Discount Amount versus Total Amount

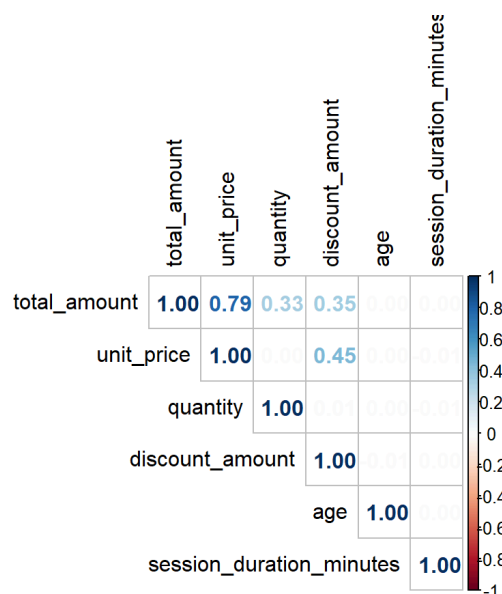


Figure 7: Correlation Matrix

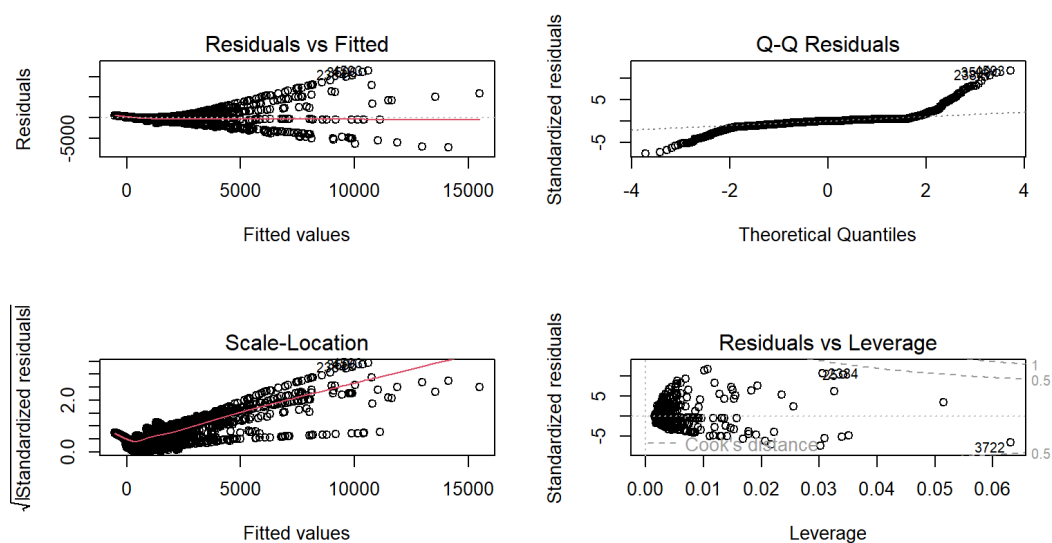


Figure 8: Diagnostic Plots for Model 3 (Raw Scale)

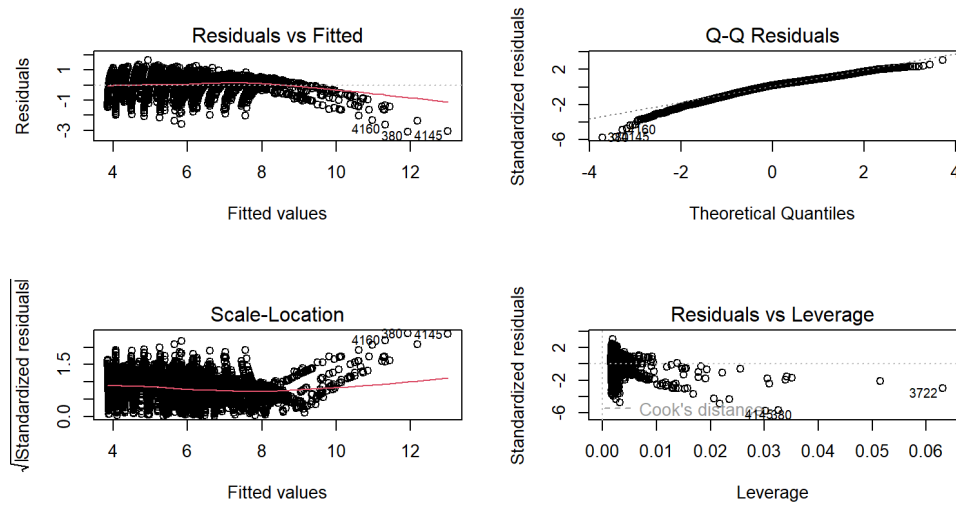


Figure 9: Diagnostic Plots for Model 4 (Log-Transformed Scale)

product_category	Mean_Spend	Median_Spend
Electronics	3732.0622	2508.195
Home & Garden	1462.7196	982.420
Sports	1131.2797	757.000
Fashion	603.2394	424.100
Toys	365.8073	238.310
Beauty	252.1493	165.760
Food	155.3129	107.920
Books	118.0918	74.365

Table 1: Average Customer Expenditure by Product Category

Model	R2	Adj R2	RMSE
Model 1 (SLR)	0.1083	0.1081	1793.05
Model 2 (Pricing MLR)	0.6259	0.6257	1161.40
Model 3 (Full MLR)	0.7366	0.7361	974.43
Model 4 (Full MLR, log)	0.8551	0.8548	0.55

Table 2: Model fit comparison across the four regression models

Hypothesis	Variable	Coefficient	P_Value	Significance	Result
H1	Quantity	0.4417	0.00e+00	Significant	Supported
H2	Unit Price	0.0009	0.00e+00	Significant	Supported
H3	Customer Age	0.0002	0.7407	Not Significant	Rejected
H4	Discount Amount	-0.0004	2.61e-05	Significant	Supported
H5	Product Category (Overall)	NA	0.00e+00	Significant	Supported
H6	Returning Customer Status	NA	0.8989	Not Significant	Rejected

Table 3: Summary of Hypothesis Table

	GVIF	Df	$GVIF^{1/(2 \cdot Df)}$
unit_price	2.480831	1	1.575065
quantity	1.002268	1	1.001133
discount_amount	1.254709	1	1.120138
age	1.001317	1	1.000658
product_category	2.249318	7	1.059611

Table 4: VIF Table